

Nirmal Patel

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PROFILE

AI/ML-focused developer with hands-on experience building real-world projects in **Deep Learning, NLP, and model deployment** using **FastAPI**. Strong foundation in **data preprocessing, feature engineering, model training & evaluation**, and deploying ML systems into usable web apps. Comfortable working across the full pipeline from dataset preparation to inference APIs while writing clean, scalable Python code.

SKILLS

Languages: Python, SQL

Frameworks & Libraries: TensorFlow, Keras, Pandas, Matplotlib, Scikit-learn, Laravel

Machine Learning & Deep Learning Concepts: Data Preprocessing, Exploratory Data Analysis (EDA), Feature Engineering, Model Training & Evaluation, Hyperparameter Tuning, Supervised & Unsupervised Learning, Classification, Regression, ANN, CNN, RNN, LSTM, NLP, Transfer Learning, Transformers

Database & Tools: MySQL, Git, VS Code, Jupyter Notebook, Postman, Hugging Face

EDUCATION

Ganpat University

Gujarat, India

B.Sc. (CA & IT) | 2022 – 2025 | CGPA: 7.97

M.Sc. (CA & IT) | 2025 – 2027 | Current CGPA: 8.98

EXPERIENCE

Backend Developer Intern - BM Coder

Jan 2025 – Jun 2025

- Designed and developed **RESTful APIs** using Laravel following MVC architecture.
- Built optimized **MySQL schemas** and improved query performance for high-usage workflows.
- Implemented **authentication, validation, error handling**, and secure CRUD operations.
- Debugged and resolved production-level issues, improving stability and reducing repeated failures.
- Worked in an Agile environment: **code reviews, sprint tasks, testing, and collaboration** with senior developers.

PROJECTS

MediScan — AI Medical Report Analyst

Sep 2025 | [GitHub](#)

- Built an end-to-end AI platform that parses blood test PDFs using a dual OCR pipeline (pdfplumber + PaddleOCR), extracts biomarkers from 6+ real lab formats, and flags abnormal values against WHO/ICMR reference ranges.
- Developed XGBoost models for liver and kidney disease risk prediction and implemented rule-based diabetes detection.
- Integrated Groq LLaMA 3 with a RAG pipeline to generate personalised, citation-grounded health explanations.
- Deployed a full-stack product with FastAPI backend, JWT auth, MySQL persistence, and React dashboard showing health score, biomarker cards, risk predictions, and report history.
- Tech:** Python, XGBoost, PaddleOCR, FastAPI, MySQL, React, Groq API, ChromaDB, LangChain, JWT, Tailwind CSS

CommentVision AI – YouTube Toxicity Analyzer

Jul 2025 | [Live Demo](#) | [GitHub](#)

- Trained a BiGRU multi-label model to detect 6 toxicity categories using the Jigsaw Toxic Comment Dataset.
- Built a FastAPI web app with Jinja2 + Chart.js to show toxicity counts, charts, and top flagged comments.
- Optimized prediction flow to process multiple comments together for faster analysis.
- Integrated YouTube comment extraction to scan up to 500 comments in <8 seconds on CPU.
- Tech:** Python, TensorFlow/Keras, Deep Learning, NLP, FastAPI, YouTube API, Chart.js

CarDamageAnalysis AI – Multi-Stage Car Damage Assessment System

Aug 2025 | [Live Demo](#) | [GitHub](#)

- Built a 3-stage EfficientNet-B3 pipeline for car detection, damage type classification, and severity grading.
- Trained on 67K+ real-world images with strong augmentation for better performance on different angles/lighting.
- Achieved strong accuracy across stages: 99.1% / 97.8% / 94.6%.
- Deployed a live app that predicts damage output with confidence scores from a single uploaded image.
- Tech:** Python, TensorFlow/Keras, CNN, EfficientNet, Transfer Learning